

Exercise 1: An Invitation to Electrodynamics

(1) By balancing the gravitational and electric forces acting on a tiny oil drop suspended between the plates of a parallel plate capacitor, Millikan observed that the following measured charges, among others, appeared at different times on a single oil drop:

$$\begin{array}{lll} 6.56 \times 10^{-19}\text{C}, & 13.13 \times 10^{-19}\text{C}, & 19.71 \times 10^{-19}\text{C}, \\ 8.20 \times 10^{-19}\text{C}, & 16.48 \times 10^{-19}\text{C}, & 22.89 \times 10^{-19}\text{C}, \\ 11.50 \times 10^{-19}\text{C}, & 18.08 \times 10^{-19}\text{C}, & 26.13 \times 10^{-19}\text{C}. \end{array}$$

What value for the elementary charge e can be deduced from these data?

(2) In a landmark experiment at the turn of the 20th century, J. J. Thomson measured an electron's charge to mass ratio e/m using a cathode ray tube and won the Nobel prize for establishing the particulate nature of the electron. (*Years later, his son G. P. Thomson won the Nobel prize for proving the wave nature of the electron!*)

In this experiment, electrons are coaxed to leave a filament by heating it, accelerated by an electric field, and deflected into a circular orbit by a magnetic field in the perpendicular direction. By measuring the accelerating potential of the electric field, the intensity of the magnetic field, and the radius of the orbit, it is possible to compute the ratio of charge to mass for the electron. Derive this expression.

Suppose you set out to reproduce Thomson's experiment with an accelerating potential of 106.2 volts, a magnetic field of 0.034 tesla, and find the radius of the orbit to be 0.001016 m, what value of e/m will you get? Using the accepted value of $1.7588 \times 10^{11}\text{C/kg}$, and the value of the electronic charge from the previous problem, estimate the mass of the electron.

(3) At the beginning of the 20th century, the idea that the rest mass of the electron might have a purely electrical origin was very attractive, especially when the equivalence of energy and mass was revealed by special relativity. Imagine the electron as a ball of charge

with radius r . Equate the potential energy of this system to mc^2 and estimate the radius of the electron. This is called the Compton radius of the electron. Although strictly not correct, you may assume, for simplicity, that the charge is concentrated at the centre in computing the potential energy. (*In a different context, Newton assumed the mass of the Earth to be concentrated at its centre in proving Kepler's laws of planetary motion. He waited twenty years to publish his results because he did not know how to justify the assumption!*)

(4) Estimate the number of coulombs of positive charge in a glass of water. Assume the volume to be half-a-litre.

(5) It is reasonable to think that we are all electrically neutral. Suppose, however, that the number of positive and negative charges in the human body differed by 0.0001% (one part in a million). With what force would two human beings at arm's length repel/attract each other? You may assume the same (reasonable) dimensions for both persons.

Can you think of an even more catastrophic consequence of such a mismatch between positive and negative charges in the human body?

Do you think that the Universe is electrically neutral? Why?

(6) Estimate the lifetime of a classical atom (before the electron loses 13 eV of energy to synchrotron radiation)?

Estimate the Bohr radius of a "gravitational hydrogen atom" *i.e.* an electron and a proton bound by their gravitational attraction alone, in its lowest energy state. How does your estimate compare with the size of the Universe?

(7) Consider an cube of side approximately 500 microns in a cubical lattice, with an electron at each lattice site. Estimate the force experienced by an electron at one such site. Calculate the initial acceleration of the charge in units of the gravitational acceleration

of the earth, g .

Compare the result with the acceleration at the surface of a black hole. The radius of the black hole may be taken to be the Schwarzschild radius $r = \frac{2GM}{c^2}$ where the mass of the black hole M can be taken to be the Chandrasekhar limit, *viz.* 1.6 times the solar mass.

(8a) A charge $q > 0$ is uniformly distributed over a thin ring of radius a . Find the electric field E on the axis of the ring as a function of the distance z from its centre.

(8b) A thin non-conducting ring of radius a is charged with a linear density $\lambda = \lambda_0 \cos \phi$ where λ_0 is a positive constant and ϕ is the azimuthal angle. Find the electric field E at its centre.

(8c) A thin straight wire filament of length $2l$ is uniformly charged by a charge q . Find the electric field E at a point at a distance x from the centre of the midpoint of the filament and located symmetrically with respect to its ends.

(8d) A semi-infinite straight uniformly charged wire has a charge λ per unit length. Find the magnitude and direction of the electric field at a point at a distance y from the wire, lying in a plane normal to the wire and passing through its end.

(9a) Find the magnetic field due to a uniform current I flowing through a long, straight wire of infinite length at an arbitrary point.

(9b) Assuming that the earth's magnetic field is due to a small current loop located at the centre of the earth, use the Biot-Savart law to calculate the magnetic moment of the small current loop, given that the field near the pole is 0.8 gauss and the radius of the earth is $R = 6 \times 10^6 \text{m}$, and $\mu_0 = 4\pi \times 10^{-7} \text{H./m}$. The magnetic moment $\vec{m}$ is defined as $\vec{m} = I\vec{A}$, where $\vec{A}$ is the area of the loop.

(9c) Find the magnetic field on the axis of circular loop due to a

uniform current I flowing through a circular loop.

(9d) Making use of the results from (10a), find the magnetic field along the axis of a solenoid having N turns per length with each turn carrying a current I . Let the point on the axis be denoted by θ_1 and θ_2 , the half-angle subtended by the ends of the solenoid at the point.

(10a) From the expression for the magnetic field due to a current in a wire of infinitesimal length $d\vec{l}$ is given by

$$d\vec{B} = \frac{I}{c} d\vec{l} \times \frac{(\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} \quad (1)$$

and the force on a circuit as if it were the sum of forces on each circuit element $d\vec{l}$

$$d\vec{F} = \frac{I}{c} d\vec{l} \times \vec{B} \quad (2)$$

show that the find the magnetic field produced by a single moving charge is

$$\vec{B}_q = q \frac{\vec{v}}{c} \times \frac{(\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} \quad (3)$$

and that the magnetic force experienced by a single moving charge is

$$\vec{F}_q = q \frac{\vec{v}}{c} \times \vec{B} \quad (4)$$

It is useful to note

$$I d\vec{l} = N q \vec{v} \quad (5)$$

where N is the total number of charges in dl .

In the presence of both electric and magnetic field, the force experienced by a test charge is called the Lorentz force, and is given by

$$\vec{F}_q = q(\vec{E} + \frac{\vec{v}}{c} \times \vec{B}) \quad (6)$$

(10b) A region is characterised by crossed electric and magnetic fields, $\vec{E} = E_0 \hat{j}$ and $\vec{B} = B_0 \hat{k}$ where E_0 and B_0 are constants. A small test charge q with mass m starts from rest at the origin at $t = 0$. Obtain the parametric equations of motion of the test charge.

(11a) Consider a circular wire loop of radius r which is being heated in such a way that the radius is a linear function of time,

$$r = vt \tag{7}$$

where v is the constant radial velocity of a point on the loop. Assume that a uniform magnetic field is present throughout the plane of the loop and is given by

$$B = B_0(1 + kt) \tag{8}$$

where B_0 and k are constants. $\vec{B}$ may be taken to be normal to the plane and directed towards the reader. Determine the induced EMF in the wire loop. Note that there are two contributions to the EMF, the first coming from the motion of the wire loop, called the motional EMF, and the second coming from the changing magnetic field, which according to equation (4) produces an induced electric field.

(11b) An electron is accelerated along a circular orbit by an increasing magnetic field which threads the orbit centrally. Show that the electron will continue to move in a path of constant radius, if the magnetic field at the path is always one-half the mean magnetic field enclosed by the path. Is the result true, if the electron is accelerated to velocities comparable with the velocity of light c .